2.6 Solving Absolute Value Inequalities

Name _____ Period ____ Date _____

Algebra I 2.6 (30-minute plan) · Student sheet

I can…

- solve absolute value inequalities of the form |ax + b| < c.
- solve absolute value inequalities of the form |ax + b| > c.
- use absolute value inequalities to solve real-life problems.

Vocabulary · fill in the missing word

Absolute value inequality — an inequality with a _____ inside absolute value bars

|x| < c — _____

|x| > c — _____

Part 1 · Notes and examples (fill in each note, then work the example)

Less than: |ax + b| < c → _____ < ax + b < _____ (“and”)

Example 1

Solve |x + 4| < 6 and graph.

Greater than: |ax + b| > c → ax + b < _____ or ax + b > _____ (“or”)

Example 2

Solve |2x − 1| ≥ 9 and graph.

Isolate first: Isolate the absolute value _____.

Example 3

Solve 3|x − 2| + 1 ≤ 13.

Part 2 · Partner practice

Partner practice

Solve $|x - 3| > 5$.